Bonus Question – Solution

(i) A vector space is a set V with a map

$$'+' : V \times V \longrightarrow V$$
$$(v_1, v_2) \longrightarrow (v_1 + v_2)$$

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(i) A vector space is a set V, with the map

(a) $'+' : V \times V \to V$

defined as $(v_1, v_2) \to (v_1 + v_2)$ $\forall\, v_1, v_2 \in V$

This is called vector addition

(b) $'\cdot' : F \times V \to V$

defined as $(\alpha, v) \to \alpha \cdot v$ $\forall\, \alpha \in F$ (a scalar from field F) and $v \in V$

(0.5)

This is called scalar multiplication.

A called F-vector space or vector space over field F if the following properties are satisfied:

(i) $(V, +)$ is an Abelian group

(ii) $\alpha \cdot (v_1 + v_2) = \alpha \cdot v_1 + \alpha \cdot v_2$ $\forall\, v_1, v_2 \in V$ and $\forall\, \alpha \in F$

(iii) $(\alpha_1 + \alpha_2) \cdot v = \alpha_1 \cdot v + \alpha_2 \cdot v$ $\forall\, v \in V$ and $\forall\, \alpha_1, \alpha_2 \in F$

(iv) $(\alpha\beta) \cdot v = \alpha(\beta v) = \beta(\alpha v)$ $\forall\, v \in V$ and $\forall\, \alpha, \beta \in F$

(v) $1 \cdot v = v$ $\forall\, v \in V$

(vi) $\alpha \cdot 0 = 0$ $\forall\, \alpha \in F$

(vii) If $v \neq 0$, then $\alpha \cdot v = 0 \implies \alpha = 0$

(viii) $0 \cdot v = 0$ $\forall\, v \in V$

(ix) If V is a vector space over a Field F, then any linear combination of vectors lying in V (with scalars from F) would again lie in V

(1.5)

(Q4) Let $V = \{x : \mathbb{R} \to \mathbb{C} \mid x \text{ is continuous and } x(t+T) = x(t) \ \forall t \in \mathbb{R}\}$

→ defining addition and scalar as usual

↳ $(x+y)(t) = x(t) + y(t)$

$(\alpha x)(t) = \alpha(x(t))$ for $x, y \in V$ and $\alpha \in \mathbb{C}$

To prove V is a vector space, proving —:

(i) $(V, +)$ is an Abelian group

→ Closure under addition

Let $x, y \in V$, Both x and y are continuous and

$x(t) = x(t+T)$, $y(t) = y(t+T)$ $\forall t$

Then

$(x+y)(t+T) = x(t+T) + y(t+T) = x(t) + y(t)$

$= (x+y)(t)$

So $x+y \in V$

→ Associativity

$((x+y)+z)(t) = (x(t) + y(t)) + z(t) = x(t) + (y(t) + z(t))$

$= (x + (y+z))(t)$

So $(x+y)+z = x + (y+z)$

→ Identity element

The zero (null) function $0 : \mathbb{R} \to \mathbb{C}$ defined by $0(t) = 0$ is continuous and $0(t+T) = 0 = 0(t)$, Hence $0 \in V$

For any $x \in V$, $(x+0)(t) = x(t) + 0 = x(t)$ so

0 is an additive identity.

→ Commutativity

Pointwise commutativity of complex addition yields

$(x+y)(t) = x(t) + y(t) = y(t) + x(t) = (y+x)(t)$

Thus, $x+y = y+x$

→ Additive Inverse

for $x \in V$, $(-x)(t) = -x(t)$, $(-x)(t+T) = -x(t+T) = -x(t)$

so $-x \in V$, $(x + (-x))(t) = x(t) - x(t) = 0$, so $x + (-x) = 0$

Thus, $(V, +)$ is an Abelian group

(2) Distributivity of scalar addition over vector

For $\alpha_1, \alpha_2 \in \mathbb{C}$ and $x \in V$

$$((\alpha_1+\alpha_2)x)(t) = (\alpha_1+\alpha_2)x(t) = \alpha_1 x(t) + \alpha_2 x(t)$$
$$= (\alpha_1 x + \alpha_2 x)(t)$$

so $(\alpha_1+\alpha_2)x = \alpha_1 x + \alpha_2 x$

(3) Distributivity of scalar over vector addition

~~$\alpha_1, \alpha_2 \in \mathbb{C}$ and $x \in V$~~ $\alpha \in \mathbb{C}$, and $x, y \in V$

$$(\alpha\cdot(x+y))(t) = \alpha\{x(t)+y(t)\} = \alpha x(t) + \alpha(y(t))$$
$$= (\alpha x + \alpha y)(t)$$

so $\alpha(x+y) = \alpha x + \alpha y$

(4) Compatibility of scalar multiplication (associativity)

for $\alpha, \beta \in \mathbb{C}$ and $x \in V$

$$((\alpha\beta)x)(t) = (\alpha\beta)x(t) = \alpha(\beta(x(t))) = (\alpha(\beta x))(t)$$

Hence $(\alpha\beta)x = \alpha(\beta x)$

(5) unit scalar

for $x \in V$, $(1\cdot x)(t) = 1\cdot x(t) = x(t)$ so $1\cdot x = x$

(6) $\alpha\cdot 0 = 0$ for all $\alpha \in \mathbb{C}$

$(\alpha\cdot 0)(t) = \alpha\cdot 0(t) = \alpha\cdot 0 = 0$ so $\alpha\cdot 0$ is a zero function

(7) $0\cdot x = 0 \;\forall\; x \in V$

$(0\cdot x)(t) = 0 \cdot x(t) = 0$

(8) if $x \neq 0$ and $\alpha\cdot x = 0$ then $\alpha = 0$

Assume $x \in V$ is not the zero function, so $\exists t_0$ with $x(t_0) \neq 0$,

if $\alpha x = 0$ (the zero function), then

$\alpha x(t_0) = 0$, Hence $\alpha x(t_0) = 0$ and $x(t_0) \neq 0$

$\longrightarrow \alpha = 0$ (since $\mathbb{C}$ is a field)

(iv) Closure under linear combinations:

Let $x_1, x_2, \ldots x_n \in V$ and $\alpha_1, \alpha_2, \ldots \alpha_n \in \mathbb{C}$

$$y(t) = \sum_{k=1}^{n} \alpha_k x_k(t)$$

Each x_k is continuous, so y is continuous.

and each x_k satisfies $x_k(t+T) = x_k(t)$

Hence

$$y(t+T) = \sum_{k=1}^{n} \alpha_k x_k(t+T) = \sum_{k=1}^{n} \alpha_k x(t) = y(t)$$

Thus, any linear combination of vectors in V lies in V. ——— (2)

∴

Since all properties are satisfied,

all continuous time signals with Time period T form a vector space.

NOTE:—

in (i), ~~writing~~ writing only some properties will fetch you only partial marks.

in (iv) for Abelian group, The given proof should be present and writing all 9 properties proofs only will fetch — (2) marks!